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> # 1st Assignment of R by
> # Tushar Sharma
> # Ref.no. - 250040208013
>
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>
>
> # ARITHMETIC OPERATORS
> # Using variables a1 and b1 for arithmetic operations
>
> a1 <- 10
> b1 <- 3
>
> print(a1 + b1)   # Addition
[1] 13
> print(a1 - b1)   # Subtraction
[1] 7
> print(a1 * b1)   # Multiplication
[1] 30
> print(a1 / b1)   # Division
[1] 3.333333
> print(a1 %% b1)  # Remainder
[1] 1
> print(a1 %/% b1) # Integer division
[1] 3
> print(a1 ^ b1)   # Power
[1] 1000
>
> # COMPARISON (RELATIONAL) OPERATORS
> # Using variables c1 and d1 for comparison operations
>
> c1 <- 8
> d1 <- 5
>
> print(c1 > d1)   # Greater than
[1] TRUE
> print(c1 < d1)   # Less than
[1] FALSE
> print(c1 == d1)  # Equal to
[1] FALSE
> print(c1 != d1)  # Not equal to
[1] TRUE
> print(c1 >= d1)  # Greater than or equal to
[1] TRUE
> print(c1 <= d1)  # Less than or equal to
[1] FALSE
>
>
> # LOGICAL OPERATORS
> # Using logical variables l1 and l2
>
> l1 <- TRUE
> l2 <- FALSE
>
> print(l1 & l2)   # Logical AND
[1] FALSE
> print(l1 | l2)   # Logical OR
[1] TRUE
> print(!l1)       # Logical NOT
[1] FALSE
>
> # ASSIGNMENT OPERATORS
> # Demonstrating different assignment operators
>
> x1 <- 5
> print(x1)
[1] 5
>
> x2 = 10
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```
> print(x2)
[1] 10
>
> 15 -> x3
> print(x3)
[1] 15
>
> #Vector used
>
> # Created a vector Apple that contain diffrent types of appel
> apple<-c('red','green','yellow','cotton')
> print(apple)
[1] "red"    "green"  "yellow" "cotton"
>
> #class Shows that what kind of data is present in the particular vector
> print(class(apple))
[1] "character"
>
> # MISCELLANEOUS OPERATORS
> # Sequence operator (:)
> print(1:5)
[1] 1 2 3 4 5
>
> #  operator (%in%)
> items <- c(2, 4, 6, 8)
> print(4 %in% items)
[1] TRUE
> print(5 %in% items)
[1] FALSE
> # VECTOR OPERATIONS
> # Creating numeric vectors and performing various operations
>
>
> v1 <- c(1, 2, 3, 4)
> v2 <- c(10, 20, 30, 40)
>
> # Arithmetic operations (element-wise)
> print(v1 + v2)   # Vector addition
[1] 11 22 33 44
> print(v1 * v2)   # Vector multiplication
[1]  10  40  90 160
> print(v2 - v1)   # Vector subtraction
[1]  9 18 27 36
> print(v2 / v1)   # Vector division
[1] 10 10 10 10
>
> # Comparison operations on vectors
> print(v1 > 2)    # Greater than comparison
[1] FALSE FALSE  TRUE  TRUE
> print(v1 <= v2)  # Vector-to-vector comparison
[1] TRUE TRUE TRUE TRUE
>
> # Mathematical functions on vectors
> print(sum(v1))       # Sum of elements
[1] 10
> print(mean(v2))      # Mean of elements
[1] 25
> print(max(v2))       # Maximum value
[1] 40
> print(min(v1))       # Minimum value
[1] 1
> print(length(v1))    # Number of elements
[1] 4
>
> # Indexing and subsetting vectors
> print(v1[1])         # First element
[1] 1
> print(v2[2:4])       # Elements from index 2 to 4
[1] 20 30 40
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> print(v1[-1])        # All elements except first
[1] 2 3 4
>
> # Conditional subsetting
> print(v2[v2 > 20])           # Elements greater than 20
[1] 30 40
> print(v1[v1 %% 2 == 0])      # Even numbers from v1
[1] 2 4
>
> # Sorting vectors
> print(sort(v2))                       # Ascending order
[1] 10 20 30 40
> print(sort(v2, decreasing = TRUE))   # Descending order
[1] 40 30 20 10
>
> # Combining and repeating vectors
> v3 <- c(v1, v2)
> print(v3)
[1]  1  2  3  4 10 20 30 40
>
> # List is an R-object which can contain different types of elements
> # like vectors, numbers, functions and even another list
> # Creating first list
>
> list_A <- list(numbers = c(10, 20, 30),name = "Tushar",passed = TRUE)
>
> print(list_A)
$numbers
[1] 10 20 30

$name
[1] "Tushar"

$passed
[1] TRUE

>
> # Creating second list
> list_B <- list(marks = c(70, 75, 80),age = 21
+ )
>
> print(list_B)
$marks
[1] 70 75 80

$age
[1] 21

>
> # Accessing elements from list
>
> print(list_A[[1]])        # Access by position
[1] 10 20 30
> print(list_A$name)        # Access by name
[1] "Tushar"
> print(list_B$marks)
[1] 70 75 80
>
> # Modifying elements in list
>
> list_A$name <- "Tushar Sharma"
> print(list_A)
$numbers
[1] 10 20 30

$name
[1] "Tushar Sharma"

$passed
```

```
[1] TRUE

>
> list_B$age <- 22
> print(list_B)
$marks
[1] 70 75 80

$age
[1] 22

>
> # Adding new elements to list
>
> list_A$grade <- "A"
> print(list_A)
$numbers
[1] 10 20 30

$name
[1] "Tushar Sharma"

$passed
[1] TRUE

$grade
[1] "A"

>
> list_B$result <- TRUE
> print(list_B)
$marks
[1] 70 75 80

$age
[1] 22

$result
[1] TRUE

>
> # Length of lists
>
> print(length(list_A))
[1] 4
> print(length(list_B))
[1] 3
>
> # Combining two lists
>
> combined_list <- c(list_A, list_B)
> print(combined_list)
$numbers
[1] 10 20 30

$name
[1] "Tushar Sharma"

$passed
[1] TRUE

$grade
[1] "A"

$marks
[1] 70 75 80

$age
[1] 22
```

```
$result
[1] TRUE

>
> # List inside list (nested list)
>
> final_list <- list(personal_details = list_A,academic_details = list_B)
>
> print(final_list)
$personal_details
$personal_details$numbers
[1] 10 20 30

$personal_details$name
[1] "Tushar Sharma"

$personal_details$passed
[1] TRUE

$personal_details$grade
[1] "A"


$academic_details
$academic_details$marks
[1] 70 75 80

$academic_details$age
[1] 22

$academic_details$result
[1] TRUE


>
> # Removing an element from list
>
> list_A$passed <- NULL
> print(list_A)
$numbers
[1] 10 20 30

$name
[1] "Tushar Sharma"

$grade
[1] "A"

>
> #Matrix is a two dimensional vector
>
> #learning how to define a matrix
> M=matrix(c('a','a','b','c','b','a'),nrow=2,ncol=3,byrow=TRUE)
> print(M)
     [,1] [,2] [,3]
[1,] "a"  "a"  "b"
[2,] "c"  "b"  "a"
>
> #Some operation on matrices
>
> M1=matrix(c(1,0,0,0,1,0,0,0,1),nrow=3,ncol=3,byrow=TRUE)
> print(M1)
     [,1] [,2] [,3]
[1,]    1    0    0
[2,]    0    1    0
[3,]    0    0    1
>
> print(dim(M1))      # Dimensions (rows, columns)
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[1] 3 3
> print(nrow(M1))     # Number of rows
[1] 3
> print(ncol(M1))     # Number of columns
[1] 3
>
>
> # Accessing matrix elements
> print(M1[1, 1])     # Element at row 1, column 1
[1] 1
> print(M1[2, ])      # Entire 2nd row
[1] 0 1 0
> print(M1[, 3])      # Entire 3rd column
[1] 0 0 1
>
> # Matrix arithmetic operations
>
> M2 <- matrix( c(2, 2, 2, 1, 1, 1,3, 3, 3),nrow = 3,ncol=3,byrow = TRUE)
>
> print(M2)
     [,1] [,2] [,3]
[1,]    2    2    2
[2,]    1    1    1
[3,]    3    3    3
>
> print(M1 + M2)      # Matrix addition
     [,1] [,2] [,3]
[1,]    3    2    2
[2,]    1    2    1
[3,]    3    3    4
> print(M2 - M1)      # Matrix subtraction
     [,1] [,2] [,3]
[1,]    1    2    2
[2,]    1    0    1
[3,]    3    3    2
> print(M1 * M2)      # Element-wise multiplication
     [,1] [,2] [,3]
[1,]    2    0    0
[2,]    0    1    0
[3,]    0    0    3
>
> # Matrix multiplication
> print(M1 %*% M2)    # Proper matrix multiplication
     [,1] [,2] [,3]
[1,]    2    2    2
[2,]    1    1    1
[3,]    3    3    3
>
>
> # Transpose of a matrix
>
> print(t(M2))
     [,1] [,2] [,3]
[1,]    2    1    3
[2,]    2    1    3
[3,]    2    1    3
>
> # Diagonal, determinant and identity check
>
> print(diag(M1))     # Diagonal elements
[1] 1 1 1
> print(det(M1))      # Determinant of matrix
[1] 1
> # FACTOR - TYPES OF SPORTS PLAYED BY STUDENTS
> sports_played <- c("Football","Handball","Volleyball","Cricket","basketb
all ","khokho")
> sport_factor <- factor(sports_played)
> print(sport_factor)
[1] Football    Handball    Volleyball  Cricket     basketball  khokho
```

```
Levels: basketball  Cricket Football Handball khokho Volleyball
>
>
> # SEMESTER 1 MARKS TABLE
> # Creating a structured table for Semester 1 marks
>
> sem1_marks <- data.frame(
+   Subject_Code = c("MDSC-101", "MDSC-102", "MDSC-103", "MDSC-104", "MDSC
-105"),
+   Subject_Name = c(
+     "Applied Linear Algebra",
+     "Inferential Statistics",
+     "Optimization Techniques",
+     "Computer Organization and Architecture",
+     "Design and Analysis of Algorithms"
+   ),
+   Grade_Point = c(8.0, 7.0, 8.0, 8.0, 9.0)
+ )
>
> print(sem1_marks)
>
> # A simple use case of the list structure
>
> student_record <- list(
+   student_name = "Tushar Sharma",
+   roll_no = 25040208013,
+   semester = "Semester 1",
+   marks_table = sem1_marks,
+   passed = TRUE
+ )
>
> print(student_record)
$student_name
[1] "Tushar Sharma"

$roll_no
[1] 25040208013

$semester
[1] "Semester 1"

$marks_table
  Subject_Code                           Subject_Name Grade_Point
1     MDSC-101                 Applied Linear Algebra           8
2     MDSC-102                 Inferential Statistics           7
3     MDSC-103                Optimization Techniques           8
4     MDSC-104 Computer Organization and Architecture           8
5     MDSC-105      Design and Analysis of Algorithms           9

$passed
[1] TRUE
```